Name - Aim - law - Theory - Tool - STEPS - Result

Name: Ohm's law

Aim: 1- achieving ohm's law Practically
2- Determine an unknown resistance value

Law: $R = \frac{V}{I}$, $V = IR$, $I = \frac{V}{R}$

$R = \frac{\rho L}{A}$, $I = \frac{Q}{T} = \frac{n.e}{T}$, $V = \frac{W}{Q}$

Theory: the current (I) through a conductor between two points is directly proportional to the voltage (V) across the two points.

Tools: battery - wires - ohm-meter voltmeter

STEPS: 1- Connect the circle
2- Change the battery voltage and take a reading of V and I
3- Repeat the previous steps and draw the relation between V and I

Name: Deflection magnetometer

Aim: Comparing the magnetic dipole moment of two bar magnets.

law: $\frac{M_1}{M_2} = \frac{\tan\theta_1}{\tan\theta_2}$

Theory: magnet is a piece of metal capable of attracting certain types of materials with a force called "magnetic force" whose effect is apparent in the magnetic field

* Gauss's first position: Compass, ruler, magnet are in a straight line
* Magnetic flux density: The amount of magnetic flux lines passing perpindicular through a unit area.
* Magnetic force: is a force resulting from the presence of a magnetic field and has a repulsive and attractive effect.
* Magnetic moment: is a measure of intensity magnetic or illustrative of magnet or any part produces a magnetic field.

Name: Photometer jolly

Aim: Comparing between the luminous intensity of two light sources.

Law: $\frac{I_1}{I_2} = \frac{d_1}{d_2}$

Tools: Two light sources - jolly photometer

Theory: luminous flex: Amount of light energy produced from a light source $\boxed{\text{unit} \rightarrow \text{lumen}}$

luminous intensity: luminous flux incident on $1m^2$ of a sphere of radius 1m (Unit → Candella)

Illuminance → luminous flux per unit area
unit → lux , $\frac{lumen}{m^2}$ ($\frac{lumen}{m^2}$)

Inverse square law → Illuminous is inversly proportional to the square of the distance from the light source

STEPS: 1- Place the two light sources on the table
2- Position the jolly photometer between them
3- Adjust the photometer until both light intensities apear equal.
4- Measure the distance from the photometer to each light source
5- Repeat with different distances and record the results in the table.

STEPS: 1- The device is set to the first Gauss Configuration
2- The distance between the magnetized needle and the magnet is constant in all cases
3- The deflection angle is measured for both magnets in all possible configuration.

Name: Concave mirror

Aim: 1- Determine of the focus length of a concave mirror.

2- Determine of the power of a concave mirror

Law: $\frac{1}{F} = \frac{1}{X} + \frac{1}{y}$

$F = \frac{100}{F}$

Theory: Center of curvature: the center of the sphere from which the mirror is a part

Mirror Pole: the mid point of the reflective surface

Radius of curvature: the distance between the centre and any point on surface

Focus point: the point where reflected rays converge

Focal length: the distance between the focus and mirror pole

Power of mirror: the ability of mirror to converge or scattering light.

Tools: Concave mirror - light source - ruler barrier.

Steps: 1- Put the body at a distance from the mirror and calculate the distance (X)

2- We move the barrier closer and farther from the light source until we take the clearest picture

3- Calculat the distance (Y) → the distance between the mirror and the barrier

4- increase the distance between the light source and the mirror gradually, take alot of values.

Name: Convex lens

Aim: 1- Determine the focal length of a convex lens

2- Determine the lens power (F)

Law: $\frac{1}{F} = \frac{1}{X} + \frac{1}{y}$

$F = \frac{100}{F}$

Theory: Center of curvature: the centre of sphere of which lens is a part

Radius of curvature: the radius of sphere of which lens is a part.

Focal point: the point at which the light rays cross.

Focal length: the distance between the focus and focal point.

lens power: the ability of lens to converges light.

Tools: Convex lens - light source - ruler - barrier.

Steps: 1- we place the body on a distance from the lens and calculate it (X)

2- Move the barrier closer and farther from the light until we take the clearest image and we calculate (y)

3- increase the distance between the object and the lens gradually and move the barrier

4- we take several reading

Name: Rectangular

Aim: Determine the refractive index of a rectangular.

Achieve: Incidentangle = Final angle.

law: $n_1 \sin\theta_1 = n_2 \sin\theta_2$, $n = \frac{\sin\theta_1}{\sin\theta_2}$

Tools: Rectangular - Piece of Paper - pins

Theory: Refraction: The deviation of light ray from it's path, as a result from it's transmission from one medium to another

Refractive index:

① absolute → it's the ratio between the speed of the light to the speed of the medium

② Relative → it's the ratio between the speed of first medium to the speed of the second medium

Steps: 1. Draw a rectangular on paper

2. Draw a line representing an incident ray θ_1

3. Place pins along this line.

4. look through the rectangular and sea the pin images. 5. Determine the direction of the exit ray by pins.

6. Determine the direction of Rectangular rays

7. Measures the angle of 1) incidence 2) refraction 3) exit

Name: Prism

Aim: To determine minimum deviation of a prism (αm)

To determine glass index of refraction (n)

law: $n_{glass} = \frac{\sin\left(\frac{A+\alpha m}{2}\right)}{\sin(A/2)}$

Tools: Prism - Piece of Paper - Pins.

Theory: 1- Reflection Phenomenon: is the change in direction of a wave front at an interface between two different media

2- Refraction Phenomenon: is the change in direction of a wave passing from one medium to another.

3- The angle of deviation: is the acute angle between the extensions of the incident and exiting rays in the prism

Steps:

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Measure the angle of 1) incidence

2) Deviation